11 The intensity of the sun's radiation, y watts per square metre, and the average distance from the sun, x astronomical units, are shown in Fig. 11 for the planets Mercury and Jupiter.

	x	y
Mercury	0.3075	14400
Jupiter	4.950	55.8

Fig. 11

The intensity y is proportional to a power of the distance x.

(i) Write down an equation for y in terms of x and two constants. **[1]**

(ii) Show that the equation can be written in the form $\ln y = a + b \ln x$. **[2]**

(iii) In the Printed Answer Booklet, complete the table for $\ln x$ and $\ln y$ correct to 4 significant figures. **[2]**

(iv) Use the values from part **(iii)** to find a and b. **[3]**

(v) Hence rewrite your equation from part **(i)** for y in terms of x, using suitable numerical values for the constants. **[2]**

(vi) Sketch a graph of the equation found in part **(v)**. **[2]**

(vii) Earth is 1 astronomical unit from the sun. Find the intensity of the sun's radiation for Earth. **[1]**

11 (i)

11 (ii)

11 (iii)

	$\ln x$	$\ln y$
Mercury		
Jupiter		

11 (iv)

11 (v)	
11 (vi)	
11(vii)	